

Long Answer Questions

Unit 1: Computer System

1. Explain the working principles of computers with a block diagram and explain it in detail.
2. Draw a neat block diagram of a Computer System showing the Input, Output, and CPU (ALU, CU, MU). Briefly explain the function of the Control Unit (CU).
3. Describe the use of computers in the education field.
4. Explain the architecture of a computer system, detailing the functions of the CPU, memory, input/output devices, and storage.
5. What is the role of an operating system in managing and coordinating the various components of a computer system with examples? Analyze the functions of an operating system.
6. Define an operating system. Explain its role in a computer system.
7. Differentiate between primary and secondary memory.
8. What is a memory unit? Write any three differences between primary and secondary memories.
9. What is a memory unit? Write any three differences between RAM and ROM.
10. Describe the role of cache memory, RAM, and storage devices. How do they contribute to overall system performance?
11. Explain the architecture of a CPU in detail.
12. Explain how the CPU interacts with both primary and secondary memory during data retrieval.
13. List the three main components of the CPU and describe their functions.
14. What are the functions of the Arithmetic Logic Unit (ALU)? Explain with examples.
15. Define registers. Write their role in CPU operation.
16. What is a Control Unit (CU)? Explain its importance in a computer system.
17. Define the motherboard and explain its primary functions.
18. What is primary memory? Describe its types briefly.
19. What is secondary memory? Give examples and explain their significance.
20. How does a scanner work? Explain the process of scanning and how software like OCR enhances the usability of scanned images.
21. What are the applications of 3D printing? Discuss its importance in fields like medicine, manufacturing, and construction.
22. What is virtual memory, and why is it important? Provide a real-life example of its use.
23. What is a device driver? Explain its importance with examples.
24. "Computers can perform different tasks depending on the programs fed to them." Justify this statement.
25. Explain the types of computer software with their respective functions.
26. Define computer software. List its main types and explain them.
27. Differentiate between system software and application software with respective roles in a computer system.

- 28.** Differentiate between open source and proprietary software models with examples of everyday applications that we use.
- 29.** Explain the concept of utility software and its role in enhancing the performance and functionality of a computer system.
- 30.** What is packaged software? How is it different from tailor-made software? Explain with examples.
- 31.** What is an operating system software? Write functions of an operating system software.

Unit 2: Number System

- 1.** No long-answer (Group C) questions on the Number System chapter were found in the source material — all its questions are short-answer conversion/operation problems (see the Short Answer document, Unit 2).

Unit 3: Block Programming

- 1.** Define block programming along with its features.
- 2.** What are the applications of block programming? Explain in brief.
- 3.** How is block programming different from traditional text-based programming?
- 4.** How does block programming make it easier for beginners to learn coding? Give examples.
- 5.** What are blocks in block programming? Describe the block components of Scratch programming language.
- 6.** Define Scratch. List and explain any two features of Scratch.
- 7.** Define scratch. Write any three features of scratch.
- 8.** List and explain the main features of Scratch programming.
- 9.** Draw and explain the main components of the Scratch interface.
- 10.** What is a sprite in Scratch? Explain its role with an example.
- 11.** Explain the purpose of the following Scratch blocks: (a) Motion block (b) Events block.
- 12.** You are given a program where the cat sprite rotates 90° and says, “Namaste” when the sprite is clicked. List and explain the block components used in this program.
- 13.** Arrange the blocks so that sprite moves to a random position, waits 1 second and then plays a sound.
- 14.** Arrange the blocks so that a sprite keeps moving 10 steps forever and bounces back when it touches the edge.
- 15.** Arrange the blocks so that a sprite asks “What is your name?”, waits for input, and then says “Hello, [name]”.
- 16.** Arrange the blocks so that a sprite plays the sound “meow” 3 times using a repeat loop.
- 17.** You want to make a Sprite move in a square pattern. Arrange the blocks to make the Sprite move 100 steps and turn 90 degrees, repeating this action 4 times.
- 18.** What is an Arduino UNO? Write down its types of components.
- 19.** What is a microcontroller? What is it used for? Why is it important in embedded systems?
- 20.** Explain any two main features of the Micro:bit device.
- 21.** Write the full form of the following: (a) MIT (b) IDE (c) PWM (d) MCV.

Unit 4: Web Technology

1. What is HTML? List out any three uses of HTML.
2. Write down the structure of HTML.
3. Explain the structure of an HTML document with an example code.
4. Explain Web Development Life Cycle.
5. List out the importance of UI/UX.
6. Define User Interface (UI) and User Experience (UX). Differentiate between them with suitable examples.
7. How is the FONT tag used? List out the attributes of the FONT tag with its attributes.
8. What are the different types of Input types in FORM?
9. What is the <DIV> tag? Write down the importance of the DIV tag.
10. What is CSS? List out its advantages.
11. Discuss the advantages and disadvantages of HTML.
12. Explain the types of CSS.
13. Differentiate between internal, inline, and external CSS with examples.
14. Explain the CSS Box Model with a neat diagram.
15. What are CSS selectors? Why is it important? Explain different types of selectors in CSS with examples.
16. Write the HTML code to create the given form (Username, Password, Login, Cancel buttons).
17. Write a HTML code to create a login form which includes User name, password input box with label and one login button and one cancel button.
18. Design an HTML registration form that includes Name (text), Email (email), Password (password), Submit and Reset buttons.
19. Write an HTML program to create a student registration form with fields: Name, Email, Password, and Submit button.
20. Design an HTML Contact Us page including: (a) Name, Email, Subject (b) Message (text area) (c) Submit button.
21. Write an HTML code to prepare your own bio-data.
22. Write an HTML code to prepare a web page about your school (introduction plus any two extracurricular activities).
23. Write HTML and CSS code to design a simple webpage with a heading, a paragraph, and a styled image.
24. Write HTML codes to display the given output (a two-column list of Computer Hardware and Computer Software).
25. Create an HTML Student Information Table to display details of 5 students including Name, Age, Class, and Grade.
26. Write an HTML program to create a table showing Student Name, Roll Number, and Marks.
27. Write the HTML code to display the given Mark-Sheet table (Name, Percentage, Division).
28. Write the HTML code to create a table with three rows and two columns showing subject name and their respective full marks. Ensure the table has a border.

29. Write HTML code to create a dropdown list of five favourite subjects.

Unit 5: Internet and Social Media

1. Define Search Engine. What are the uses of search engines? Write with an example.
2. Explain the concept of Remote Login.
3. Write down the difference between HTTP and HTTPS.
4. What is social media? List out the uses of social media with its examples.
5. Define blogs. Write down the advantages and disadvantages of blogging.
6. Write any four/four social media advantages and disadvantages.
7. What are email security solutions? Explain their components and importance.

Unit 6: Cyber Security and Digital Citizenship

1. Define malware and describe its types.
2. What is malware? Write down the different types of malwares. (see also Unit 1 short-answer list)
3. Explain about digital footprint and write the advantages and disadvantages of it.
4. Define digital footprint. Write two-two differences between active and passive digital footprint.
5. Define digital footprint. Write three guidelines of being a good digital citizen.
6. Explain the safe browsing techniques.
7. Describe safe web browsing techniques. How do they help prevent cybercrime?
8. What are secure websites? Write any three ways to check if a website is secure or not.
9. Define Cybercrime. Explain the prevention method from cybercrime.
10. What is cybercrime? Describe its effects on individuals, businesses, and society.
11. Explain any four major types of cybercrime with examples.
12. Explain the methods of preventing cybercrime.
13. Define cybercrime. Explain any three safe browsing techniques.
14. Explain about netiquette guidelines.
15. What do you mean by netiquette? Mention three good online behaviors.
16. Define digital citizenship. Explain the role of netiquette and online behavior in becoming a responsible digital citizen.
17. What is cyber security? Why is it important for individuals and businesses?
18. What is cyber security? Write down preventive measures to protect yourself from cyber threats.
19. Define cybersecurity. Explain its importance in today's digital world.
20. What is a firewall? Explain three web browsing practices to minimize cyber threats.
21. Explain the importance of using strong passwords.
22. Write four rules to create a strong password.
23. Define password managers. Explain their features and advantages.
24. Describe the role of VPN in preventing cybercrime.
25. What is security awareness training? Write any four advantages of it.

26. What is cyberbullying? Mention any three preventive measures against cyberbullying.

Unit 7: Concept of Programming

- 1.** Explain programming languages with its types.
- 2.** Differentiate between compiler and interpreter.
- 3.** Define algorithm and flowchart.
- 4.** Write five symbols with their functions of flowchart.
- 5.** Create an algorithm to find whether the given number is odd or even. Also make flowchart from the algorithm.
- 6.** Explain Python programming with its features.
- 7.** Describe the use of input/output statements in Python.
- 8.** Define formatted string literals. Demonstrate its use in a simple program where the user inputs their name and the output is displayed "Hello (user_name)! Welcome to our program".
- 9.** Explain the data types with examples.
- 10.** Describe typecasting with its types.
- 11.** List the types of operators. Explain any one of them in detail with examples.
- 12.** Explain a conditional statement with an example.
- 13.** Define iteration. Differentiate for and while loop.
- 14.** Differentiate between list and dictionary.
- 15.** Given is the list elements: [Kathmandu, Biratnagar, Nepalgunj, Pokhara, Butwal, Birendranagar, Mahendranagar, Balefi] — (a) Add list item Malangawa. (b) Remove list item Balefi. (c) Print the list items one by one using a loop.
- 16.** Write a Python program that takes the monthly unit consumption as input and calculates the total bill based on the given criteria (0–20 units: Rs 30 minimum; 21–45 units: Rs 5/unit; above 45 units: Rs 7/unit).
- 17.** Write a Python program to check whether a number is positive, negative, or zero using if...elif...else statements.
- 18.** Write a Python program to input two numbers from the user and display their sum, difference, product and quotient.
- 19.** Write a Python program that checks if a given character is a vowel or not (a, e, i, o, u).
- 20.** Write a Python program to input a number n and print the first n Fibonacci numbers using a loop.
- 21.** Write a Python program to display the Fibonacci series up to 10 terms.
- 22.** Write a Python program to input a number and print whether it is an Armstrong number or not.
- 23.** Write a Python program to input a list of numbers and print the second largest number.
- 24.** Write a Python program to input a number and print its factorial using a function.
- 25.** Write a program to calculate the factorial of a number using a loop.
- 26.** Write a Python program to input a string and check whether it is a palindrome.
- 27.** Write a Python program to check whether a given string is a palindrome.
- 28.** Write a Python program to input a grade in character and then print the appropriate message as given (A+, A, B+, B, C+, C).

- 29.** Write a python program to input three different numbers and find the largest one.
- 30.** Write a Python program to input three numbers and find the greatest among them.
- 31.** Write a Python program that takes a student's score as input and displays a comment based on the given 3 categories, using an if-elif-else structure and a formatted print statement.
- 32.** Write a Python program to calculate the sum, difference, product, and average of two numbers.
- 33.** Write a Python program to display the multiplication table of a given number.
- 34.** Write a program to find the sum of all even numbers between 1 and 100.
- 35.** Write a Python program to reverse a given string.
- 36.** Write a program to calculate the square and cube of a number.
- 37.** Write a Python program to calculate the sum of the digits of a given number.